

FAYETTE REGIONAL, TX, USA

WMO: 722554

Lat: 29.908N

Lon: 96.950W

Elev: 324

StdP: 14.52

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 12978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.2	32.5	12.1	10.3	48.3	18.4	14.1	48.1	21.8	58.5	19.7	56.0	3.7	0	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.8	99.7	75.3	98.0	75.5	96.0	76.1	80.9	88.3	80.2	88.1	79.5	87.9	7.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	154.2	82.8	78.5	149.6	82.6	77.5	144.8	82.3	44.7	88.7	43.9	88.2	43.2	87.6	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.5	14.3	25.1	102.8	4.0	2.8	22.3	104.9	19.9	106.5	17.7	108.1	14.8	110.2		
			23.5	82.0	4.1	2.1	20.5	83.6	18.1	84.8	15.8	86.0	12.9	87.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.0	52.5	56.2	63.3	69.5	76.7	82.8	84.5	85.7	80.6	71.6	61.9	54.1
	DBStd	13.73	9.34	10.19	8.53	6.81	5.55	3.28	2.93	3.11	4.49	7.73	9.07	9.38
	HDD50	214	80	50	9	1	0	0	0	0	0	0	13	61
	HDD65	1400	401	277	135	34	3	0	0	0	0	31	162	357
	CDD50	7530	158	223	422	585	828	985	1071	1108	919	671	371	189
	CDD65	3240	14	29	83	169	366	535	606	643	469	236	70	20
	CDH74	34477	58	143	430	1255	3445	6156	7303	8103	4841	2177	466	100
	CDH80	16105	2	18	60	307	1321	3038	3790	4456	2263	761	79	10
(14) Wind	WSAvg	5.4	5.9	6.5	6.5	6.7	6.2	5.2	4.6	4.3	3.5	4.3	5.2	5.5
(15) Precipitation	PrecAvg	39.1	3.2	2.7	2.8	3.2	3.9	4.2	2.5	2.4	3.5	4.8	3.7	3.1
	PrecMax	54.5	8.8	8.0	7.2	9.1	11.0	10.5	7.2	6.3	7.6	13.5	12.0	9.6
	PrecMin	21.5	0.1	0.2	0.3	0.4	0.2	0.0	0.1	0.3	0.3	0.0	0.4	0.9
	PrecStd	10.4	2.1	1.9	1.6	2.3	2.8	2.6	2.0	1.8	2.4	4.0	3.1	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	82.4	84.5	90.1	94.8	100.4	100.8	102.5	99.4	93.0	85.7	81.2
		MCWB	65.1	67.4	69.2	70.9	74.7	75.1	75.4	75.9	72.2	74.4	71.6	68.6
	2%	DB	75.0	78.6	81.5	86.2	91.3	97.3	98.8	100.2	96.6	90.2	82.0	76.8
		MCWB	62.6	65.6	68.0	70.8	74.6	75.2	75.4	76.1	74.3	74.5	69.7	65.8
	5%	DB	72.1	74.8	79.1	83.5	90.0	94.9	96.8	98.8	93.2	87.7	79.3	73.3
		MCWB	63.7	65.2	67.2	70.7	74.5	76.3	76.2	75.8	75.3	73.2	68.1	65.3
	10%	DB	69.1	72.0	75.8	81.2	87.7	92.5	94.7	96.7	90.8	84.4	76.6	71.1
		MCWB	61.5	65.2	66.5	68.7	74.2	76.7	77.0	76.2	74.6	71.1	67.5	64.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	71.9	73.4	77.2	79.9	82.2	81.5	81.4	81.1	79.9	76.0	73.1
		MCDB	73.9	76.7	79.4	83.4	87.8	88.6	88.7	88.7	86.6	84.5	81.1	76.6
	2%	WB	68.1	70.1	71.9	75.1	78.4	80.6	80.5	80.5	79.8	78.2	73.3	70.6
		MCDB	72.8	74.7	77.1	81.1	85.9	88.4	88.5	88.7	86.6	84.1	77.8	74.8
	5%	WB	65.9	68.2	70.7	73.7	77.3	79.7	79.8	79.8	78.9	76.9	71.4	68.4
		MCDB	70.6	73.2	75.3	79.8	84.8	87.9	88.4	88.8	86.1	83.2	76.6	72.1
	10%	WB	63.3	66.0	69.1	72.3	76.2	78.7	79.1	79.1	78.0	75.3	69.6	65.2
		MCDB	67.0	70.5	73.6	78.1	83.3	86.7	87.8	88.3	85.4	81.4	74.3	69.6
(35) Mean Daily Temperature Range	5% DB	MDBR	20.2	19.5	19.9	20.1	18.6	18.6	19.2	20.8	19.5	21.4	20.8	20.8
		MCDBR	23.5	22.0	20.1	21.0	20.3	22.1	22.6	23.5	22.4	21.3	21.6	23.4
	5% WB	MCWBR	14.2	12.8	9.9	9.9	7.1	5.6	5.0	5.0	6.5	8.9	11.9	15.8
		MCWBR	18.0	18.4	16.4	16.4	16.0	17.1	18.4	20.5	17.4	17.0	17.8	20.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.337	0.343	0.368	0.412	0.463	0.441	0.460	0.449	0.428	0.377	0.371	0.327
		taud	2.469	2.468	2.399	2.293	2.193	2.307	2.285	2.304	2.320	2.445	2.396	2.513
	Ebn at Noon	Ebn at Noon	284	293	291	280	265	270	264	266	269	277	269	282
		Edh at Noon	30	32	37	42	47	42	43	41	39	32	31	27
(44) All-Sky Solar Radiation	RadAvg	RadAvg	927	1058	1399	1683	1838	2038	2052	1942	1624	1363	1019	822
	RadStd	RadStd	97	196	126	141	140	138	169	108	153	171	114	104

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+158

Nomenclature: See separate page